Q1. [23 pts] True/False

(a) [1 pt] Solving a non linear separation problem with a hard margin Kernelized SVM (Gaussian RBF Kernel) might lead to overfitting.

● True ○ False

(b) [1 pt] In SVMs, the sum of the Lagrange multipliers corresponding to the positive examples is equal to the sum of the Lagrange multipliers corresponding to the negative examples.

● True ○ False

(c) [1 pt] SVMs directly give us the posterior probabilities $P(y = 1|x)$ and $P(y = -1|x)$.

○ True ● False

(d) [1 pt] $V(X) = E[X]^2 - E[X^2]$

○ True ● False

(e) [1 pt] In the discriminative approach to solving classification problems, we model the conditional probability of the labels given the observations.

● True ○ False

(f) [1 pt] In a two class classification problem, a point on the Bayes optimal decision boundary x^* always satisfies $P(y = 1|x^*) = P(y = 0|x^*)$.

○ True ● False

(g) [1 pt] Any linear combination of the components of a multivariate Gaussian is a univariate Gaussian.

● True ○ False

(h) [1 pt] For any two random variables $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$, $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

○ True ● False

(i) [1 pt] Stanford and Berkeley students are trying to solve the same logistic regression problem for a dataset. The Stanford group claims that their initialization point will lead to a much better optimum than Berkeley's initialization point. Stanford is correct.

○ True ● False

(j) [1 pt] In logistic regression, we model the odds ratio $(\frac{p}{1-p})$ as a linear function.

○ True ● False

(k) [1 pt] Random forests can be used to classify infinite dimensional data.

● True ○ False

(l) [1 pt] In boosting we start with a Gaussian weight distribution over the training samples.

○ True ● False

(m) [1 pt] In Adaboost, the error of each hypothesis is calculated by the ratio of misclassified examples to the total number of examples.

○ True ● False

(n) [1 pt] When $k = 1$ and $N \to \infty$, the kNN classification rate is bounded above by twice the Bayes error rate.

● True ○ False

(o) [1 pt] A single layer neural network with a sigmoid activation for binary classification with the cross entropy loss is exactly equivalent to logistic regression.

● True ○ False

(p) [1 pt] The loss function for LeNet5 (the convolutional neural network by LeCun et al.) is convex.

○ True ● False

(q) [1 pt] Convolution is a linear operation i.e. $(\alpha f_1 + \beta f_2) * g = \alpha f_1 * g + \beta f_2 * g$.

● True ○ False

(r) [1 pt] The k-means algorithm does coordinate descent on a non-convex objective function.

● True ○ False

(s) [1 pt] A 1-NN classifier has higher variance than a 3-NN classifier.

● True ○ False

(t) [1 pt] The single link agglomerative clustering algorithm groups two clusters on the basis of the maximum distance between points in the two clusters.

○ True ● False

(u) [1 pt] The largest eigenvector of the covariance matrix is the direction of minimum variance in the data.

○ True ● False

(v) [1 pt] The eigenvectors of AA^T and A^TA are the same.

○ True ● False

(w) [1 pt] The non-zero eigenvalues of AA^T and A^TA are the same.

● True ○ False

Q2. [36 pts] Multiple Choice Questions

(a) [4 pts] In linear regression, we model $P(y|x) \sim \mathcal{N}(w^T x + w_0, \sigma^2)$. The irreducible error in this model is _____.

- [x] σ^2
- [ ] $E[(y - E[y|x])|x]$
- [x] $E[(y - E[y|x])^2|x]$
- [ ] $E[y|x]$

(b) [4 pts] Let S_1 and S_2 be the set of support vectors and w_1 and w_2 be the learnt weight vectors for a linearly separable problem using hard and soft margin linear SVMs respectively. Which of the following are correct?

- [ ] $S_1 \subset S_2$
- [x] S_1 may not be a subset of S_2
- [ ] $w_1 = w_2$
- [x] w_1 may not be equal to w_2.

(c) [4 pts] Ordinary least-squares regression is equivalent to assuming that each data point is generated according to a linear function of the input plus zero-mean, constant-variance Gaussian noise. In many systems, however, the noise variance is itself a positive linear function of the input (which is assumed to be non-negative, i.e., $x \geq 0$). Which of the following families of probability models correctly describes this situation in the univariate case?

- [x] $P(y|x) = \frac{1}{\sigma\sqrt{2\pi x}}\exp(-\frac{(y-(w_0+w_1x))^2}{2x\sigma^2})$
- [ ] $P(y|x) = \frac{1}{\sigma\sqrt{2\pi x}}\exp(-\frac{(y-(w_0+(w_1+\sigma^2)x))^2}{2\sigma^2})$
- [ ] $P(y|x) = \frac{1}{\sigma\sqrt{2\pi}}\exp(-\frac{(y-(w_0+w_1x))^2}{2\sigma^2})$
- [ ] $P(y|x) = \frac{1}{\sigma x\sqrt{2\pi}}\exp(-\frac{(y-(w_0+w_1x))^2}{2x^2\sigma^2})$

(d) [3 pts] The left singular vectors of a matrix A can be found in _____.

- [x] Eigenvectors of AA^T
- [ ] Eigenvectors of A^2
- [ ] Eigenvectors of A^TA
- [ ] Eigenvalues of AA^T

(e) [3 pts] Averaging the output of multiple decision trees helps _____.

- [ ] Increase bias
- [ ] Increase variance
- [ ] Decrease bias
- [x] Decrease variance

(f) [4 pts] Let A be a symmetric matrix and S be the matrix containing its eigenvectors as column vectors, and D a diagonal matrix containing the corresponding eigenvalues on the diagonal. Which of the following are true:

- [x] $AS = SD$
- [ ] $SA = DS$
- [ ] $AS = DS$
- [ ] $AS = DS^T$

(g) [4 pts] Consider the following dataset: $A = (0, 2)$, $B = (0, 1)$ and $C = (1, 0)$. The k-means algorithm is initialized with centers at A and B. Upon convergence, the two centers will be at

- [ ] A and C
- [ ] C and the midpoint of AB
- [x] A and the midpoint of BC
- [ ] A and B

(h) [3 pts] Which of the following loss functions are convex?

- ◯ Misclassification loss
- ● Hinge loss
- ● Logistic loss
- ● Exponential Loss ($e^{(-yf(x))}$)

(i) [3 pts] Consider T_1, a decision stump (tree of depth 2) and T_2, a decision tree that is grown till a maximum depth of 4. Which of the following is/are correct?

- ◯ $Bias(T_1) < Bias(T_2)$
- ● $Variance(T_1) < Variance(T_2)$
- ● $Bias(T_1) > Bias(T_2)$
- ◯ $Variance(T_1) > Variance(T_2)$

(j) [4 pts] Consider the problem of building decision trees with k-ary splits (split one node into$k$ nodes) and you are deciding k for each node by calculating the entropy impurity for different values of k and optimizing simultaneously over the splitting threshold(s) and k. Which of the following is/are true?

- ◯ The algorithm will always choose $k = 2$
- ● There will be $k-1$ thresholds for a k-ary split
- ● The algorithm will prefer high values of k
- ◯ This model is strictly more powerful than a binary decision tree.

Q1. [9 pts] True or False

(a) [1 pt] The singular value decomposition of a real matrix is unique.

○ True ● False

(b) [1 pt] A multiple-layer neural network with linear activation functions is equivalent to one single-layer perceptron that uses the same error function on the output layer and has the same number of inputs.

● True ○ False

(c) [1 pt] The maximum likelihood estimator for the parameter θ of a uniform distribution over $[0, \theta]$ is unbiased.

○ True ● False

(d) [1 pt] The k-means algorithm for clustering is guaranteed to converge to a local optimum.

● True ○ False

(e) [1 pt] Increasing the depth of a decision tree cannot increase its training error.

● True ○ False

(f) [1 pt] There exists a one-to-one feature mapping ϕ for every valid kernel k.

○ True ● False

(g) [1 pt] For high-dimensional data, k-d trees can be slower than brute force nearest neighbor search.

● True ○ False

(h) [1 pt] If we had infinite data and infinitely fast computers, kNN would be the only algorithm we would study in CS 189.

● True ○ False

(i) [1 pt] For datasets with high label noise (many data points with incorrect labels), random forests would generally perform better than boosted decision trees.

● True ○ False

Q2. [24 pts] Multiple Choice

(a) [2 pts] In Homework 4, you fit a logistic regression model on spam and ham data for a Kaggle Competition. Assume you had a very good score on the public test set, but when the GSIs ran your model on a private test set, your score dropped a lot. This is likely because you overfitted by submitting multiple times and changing the following between submissions:

- ● λ, your penalty term
- ● ϵ, your convergence criterion
- ● η, your step size
- ● Fixing a random bug

(b) [2 pts] Given d-dimensional data $\{\mathbf{x}_i\}_{i=1}^N$, you run principle component analysis and pick P principle components. Can you always reconstruct any data point $\mathbf{x}_i$ for $i \in \{1 \dots N\}$ from the P principle components with zero reconstruction error?

- ○ Yes, if $P < d$
- ● Yes, if $P = d$
- ○ Yes, if $P < n$
- ○ No, always

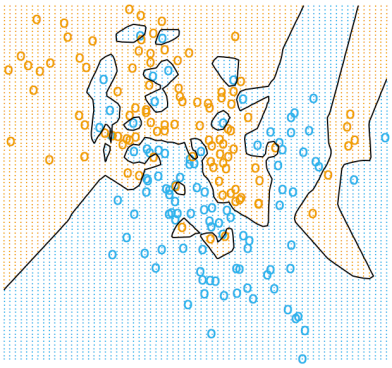
(c) [2 pts] Putting a standard Gaussian prior on the weights for linear regression ($\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$) will result in what type of posterior distribution on the weights?

- ○ Laplace
- ○ Uniform
- ○ Poisson
- ● None of the above

(d) [2 pts] Suppose we have N instances of d-dimensional data. Let h be the amount of data storage necessary for a histogram with a fixed number of ticks per axis, and let k be the amount of data storage necessary for kernel density estimation. Which of the following is true about h and k?

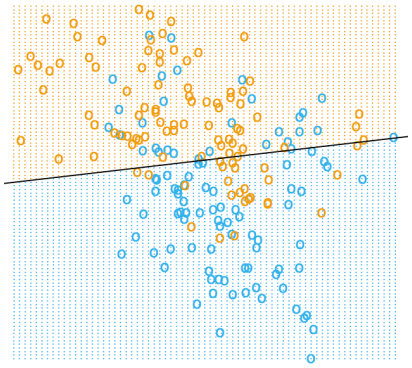
- ○ h and k grow linearly with N
- ● h grows exponentially with d, and k grows linearly N
- ○ h and k grow exponentially with d
- ○ h grows linearly with N, and k grows exponentially with d

(e) [2 pts] Which of the these classifiers could have generated this decision boundary?



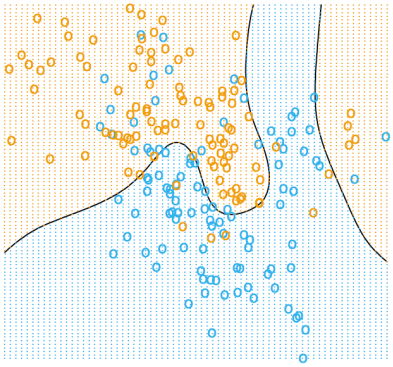
- ○ Linear SVM
- ● 1-NN
- ○ Logistic regression
- ○ None of the above

(f) [2 pts] Which of the these classifiers could have generated this decision boundary?



- [x] Linear SVM
- [ ] 1-NN
- [x] Logistic regression
- [ ] None of the above

(g) [2 pts] Which of the these classifiers could have generated this decision boundary?



- [ ] Linear SVM
- [ ] 1-NN
- [ ] Logistic regression
- [x] None of the above

(h) [2 pts] You want to cluster this data into 2 clusters. Which of the these algorithms would work well?



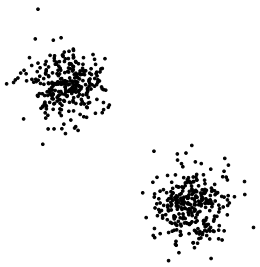
- [ ] K-means
- [x] GMM clustering
- [ ] Mean shift clustering

(i) [2 pts] You want to cluster this data into 2 clusters. Which of the these algorithms would work well?



○ K-means ○ GMM clustering ● Mean shift clustering

(j) [2 pts] You want to cluster this data into 2 clusters. Which of the these algorithms would work well?



● K-means ● GMM clustering ● Mean shift clustering

The following questions are about how to help CS 189 TA Jonathan Snow to solve the homework.

(k) [2 pts] Jonathan just trained a decision tree for a digit recognition. He notices an extremely low training error, but an abnormally large test error. He also notices that an SVM with a linear kernel performs much better than his tree. What could be the cause of his problem?

● Decision tree is too deep

● Decision tree is overfitting

○ Learning rate too high

○ There is too much training data

(l) [2 pts] Jonathan has now switched to multilayer neural networks and notices that the training error is going down and converges to a local minimum. Then when he tests on the new data, the test error is abnormally high. What is probably going wrong and what do you recommend him to do?

● The training data size is not large enough. Collect a larger training data and retrain it.

● Play with learning rate and add regularization term to the objective function.

● Use a different initialization and train the network several times. Use the average of predictions from all nets to predict test data.

○ Use the same training data but add two more hidden layers.

Q1. [44 pts] True or False

(a) [2 pts] A neural network with multiple hidden layers and sigmoid nodes can form non-linear decision boundaries.

● True ○ False

(b) [2 pts] All neural networks compute non-convex functions of their parameters.

○ True ● False

(c) [2 pts] For logistic regression, with parameters optimized using a stochastic gradient method, setting parameters to 0 is an acceptable initialization.

● True ○ False

(d) [2 pts] For arbitrary neural networks, with weights optimized using a stochastic gradient method, setting weights to 0 is an acceptable initialization.

○ True ● False

(e) [2 pts] Given a design matrix $X \in \mathbb{R}^{n \times d}$, where $d \ll n$, if we project our data onto a k dimensional subspace using PCA where k equals the rank of X, we recreate a perfect representation of our data with no loss.

● True ○ False

(f) [2 pts] Hierarchical clustering methods require a predefined number of clusters, much like k-means.

○ True ● False

(g) [2 pts] Given a predefined number of clusters k, globally minimizing the k-means objective function is NP-hard.

● True ○ False

(h) [2 pts] Using cross validation to select hyperparameters will guarantee that our model does not overfit.

○ True ● False

(i) [2 pts] A random forest is an ensemble learning method that attempts to lower the bias error of decision trees.

○ True ● False

(j) [2 pts] Bagging algorithms attach weights $w_1 ... w_n$ to a set of N weak learners. They re-weight the learners and convert them into strong ones. Boosting algorithms draw N sample distributions (usually with replacement) from an original data set for learners to train on.

○ True ● False

(k) [2 pts] Given any matrix X, its singular values are the eigenvalues of $XX^\top$ and $X^\top X$.

○ True ● False

(l) [2 pts] Given any matrix X, $(XX^\top + \lambda I)^{-1}$ for $\lambda \neq 0$ always exists.

○ True ● False

(m) [2 pts] Backpropagation is motivated by utilizing Chain Rule and Dynamic Programming to conserve mathematical calculations.

● True ○ False

(n) [2 pts] An infinite depth binary Decision Tree can always achieve 100% training accuracy, provided that no point is mislabeled in the training set.

● True ○ False

(o) [2 pts] In One vs All Multi-Class Classification in SVM, we are trying to classify an input data point X as one of the N classes ($C_1 ... C_n$), each of which has a parameter vector $\vec{w}_1 ... \vec{w}_n$. We classify point X as the class C_i which maximizes the inner product of X and $\vec{w}_i$. ● True ○ False

(p) [2 pts] The number of parameters in a parametric model is fixed, while the number of parameters in a non-parametric model grows with the amount of training data.

● True ○ False

(q) [2 pts] As model complexity increases, bias will decrease while variance will increase.

● True ○ False

(r) [2 pts] Consider a cancer diagnosis classification problem where almost all of the people being diagnosed don't have cancer. The probability of correct classification is the most important metric to optimize.

○ True ● False

(s) [2 pts] For the 1-Nearest Neighbors algorithm, as the number of data points increases to infinity in our dataset, the error of our algorithm is guaranteed to be bounded by twice the Bayes Risk.

● True ○ False

(t) [2 pts] Increasing the dimensionality of our data always decreases our misclassification rate.

○ True ● False

(u) [2 pts] It is possible to represent a XOR function with a neural network without a hidden layer.

○ True ● False

(v) [2 pts] At high dimensionality, the KD tree speedup to the nearest neighbor can be slower than the naive nearest neighbor implementation.

● True ○ False

Q2. [33 pts] Multiple Choice

(a) [3 pts] Given a Neural Net with N input nodes, no hidden layers, one output node, with Entropy Loss and Sigmoid Activation Functions, which of the following algorithms (with the proper hyper-parameters and initialization) can be used to find the global optimum?

- ○ Simulated Annealing (Gradient Descent with restarts)
- ○ Stochastic Gradient Descent
- ○ Mini-Batch Gradient Descent
- ○ Batch Gradient Descent
- ● All of the above
- ○ None of the above

(b) [3 pts] Given function $f(x) = |x^2 + 3| - 1$ defined on $\mathbb{R}$:

- ● Newtons Method on minimizing gradients will always converge to the global optimum in one iteration from any starting location
- ○ Stochastic Gradient Descent will always converge to the global optimum in one iteration
- ○ The problem is nonconvex, so it not feasible to find a solution.
- ○ All of the above
- ○ None of the above

(c) [3 pts] Daniel wants to minimize a convex loss function f(x) using stochastic gradient descent. Given a random starting point, mark the condition that would guarantee that stochastic gradient descent will converge to the global optimum. Let η_t = step size at iteration t.

- ○ $\eta_t < 0$
- ○ Constant step size η_t
- ● Decreasing step size $\eta_t = \frac{1}{\sqrt{t}}$
- ○ Decreasing step size $\eta_t = \frac{1}{t^2}$
- ○ All of the above
- ○ None of the above

(d) [3 pts] Which of the following is true of logistic regression?

- ○ It can be motivated by "log odds"
- ○ The optimal weight vector can be found using MLE.
- ○ It can be used with L1 regularization
- ● All of the above
- ○ None of the above

(e) [3 pts] You've just finished training a decision tree for spam classification, and it is getting abnormally bad performance on both your training and test sets. You know that your implementation has no bugs, so what could be causing the problem?

- ● Your decision trees are too shallow.
- ○ You need to increase the learning rate.
- ○ You are overfitting.
- ○ All of the above.

(f) [3 pts] The numerical output of a sigmoid node in a neural network:

- ○ Is unbounded, encompassing all real numbers.
- ○ Is unbounded, encompassing all integers.
- ● Is bounded between 0 and 1.
- ○ Is bounded between -1 and 1.

(g) [3 pts] If n is the number of points in the training set, regular nearest neighbor (without KD trees, hashing, etc) has a classification runtime of:

- ○ $O(1)$
- ● $O(n)$
- ○ $O(\log n)$
- ○ $O(n^2)$

(h) [3 pts] Consider the p-norm of a vector x defined using the notation $\|x\|_p$. Also note that α is a scalar. Which of the following is true?

- ○ $\|x\|_p + \|y\|_p \geq \|x + y\|_p$.
- ○ $\|x\|_p = 0$ implies x is the zero vector.
- ○ $\|\alpha x\|_p = |\alpha| \|x\|_p$.
- ● All of the above.

(i) [3 pts] What are some practical problems with the sigmoidal activation function in neural nets?

- ○ It is convex, and convex functions cannot solve nonconvex problems
- ○ It can have negative values
- ○ It does not work well with the entropy loss function
- ● Gradients are small for values away from 0, leading to the "Vanishing Gradient" problem for large or recurrent neural nets

(j) [3 pts] In Homework 4, you fit a logistic regression model on spam and ham data for a Kaggle Competition. Assume you had a very good score on the public test set, but when the GSIs ran your model on a private test set, your score dropped a lot. This is likely because you overfitted by submitting multiple times and changing which of the following between submissions: A) λ, your penalty term; B) η, your step size; C) ϵ, your convergence criterion; or D) Fixing a random bug:

- ○ A
- ○ A, B, and C
- ○ B
- ○ C and D
- ○ A and B
- ● A, B, C, and D

(k) [3 pts] With access to an n-by-n matrix of pairwise data distances, but no access to the data itself, we can use which of the following clustering techniques: A) k-means; B) k-medoids; C) hierarchical clustering:

- ○ A
- ○ A and B
- ○ B
- ● B and C
- ○ C
- ○ A, B, and C

(l) [0 pts] What was your favorite class of the ~~semester~~ ~~year~~ all time?

- ● CS 189 - Introduction to Machine Learning
- ● CS 189 - Kaggle Competitions for Dummies
- ● CS 189 - Classify EVERYTHING
- ● All of the above
- ● CS 189 - Advanced MATLAB and Numpy
- ● ~~None of the above~~ (Choose this if you dare...)

Q1. [30 pts] True or False

(1) [2 pts] Random forests usually perform better than AdaBoost when your dataset has mislabeled data points.
● True ○ False

(2) [2 pts] The discriminant function computed by kernel methods are a linear function of its parameters, not necessarily a linear function of the inputs.
● True ○ False

(3) [2 pts] The XOR operator can be modeled using a neural network with a single hidden layer (i.e. 3-layer network).
● True ○ False

(4) [2 pts] Convolutional neural networks are rotation invariant.
○ True ● False

(5) [2 pts] Making a decision tree deeper will assure better fit but reduce robustness.
● True ○ False

(6) [2 pts] Bagging makes use of the bootstrap method.
● True ○ False

(7) [2 pts] K-means automatically adjusts the number of clusters.
○ True ● False

(8) [2 pts] Dimensionality reduction can be used as pre-processing for machine learning algorithms like decision trees, kd-trees, neural networks etc.
● True ○ False

(9) [2 pts] K-d trees guarantee an exponential reduction in the time it takes to find the nearest neighbor of an example as compared to the naive method of comparing the distances to every other example.
○ True ● False

(10) [2 pts] Logistic regression is equivalent to a neural network without hidden units and using cross-entropy loss.
● True ○ False

(11) [2 pts] Convolutional neural networks generally have fewer free parameters as compared to fully connected neural networks.
● True ○ False

(12) [2 pts] K-medoids is a kind of agglomerative clustering.
○ True ● False

(13) [2 pts] Whitening the data doesn't change the first principal direction.
○ True ● False

(14) [2 pts] PCA can be kernelized.
● True ○ False

(15) [2 pts] Performing K-nearest neighbors with $K = N$ yields more complex decision boundaries than 1-nearest neighbor.
○ True ● False

Q2. [30 pts] Multiple Choice

(1) [3 pts] Which of the following guidelines is applicable to initialization of the weight vector in a fully connected neural network.

- ◯ Should not set it to zero since otherwise it will cause overfitting
- ◯ Should set it to zero since otherwise it causes a bias
- ● Should not set it to zero since otherwise (stochastic) gradient descent will explore a very small space
- ◯ Should set it to zero in order to preserve symmetry across all neurons

(2) [3 pts] Duplicating a feature in linear regression

- ● Can reduce the L2-Penalized Residual Sum of Squares.
- ◯ Can reduce the L1-Penalized Residual Sum of Squares (RSS).
- ● Does not reduce the Residual Sum of Squares (RSS).
- ◯ None of the above

(3) [3 pts] Which of the following is/are forms of regularization in neural networks.

- ● Weight decay
- ● L1 regularization
- ● L2 regularization
- ● Dropout

(4) [3 pts] We are given a classifier that computes probabilities for two classes (positive and negative). The following is always true about the ROC curve, and the area under the ROC curve (AUC):

- ◯ An AUC of 0.5 represents a classifier that performs worse than random.
- ● The ROC curve allows us to visualize the tradeoff between true positive and false positive classifications.
- ● We generate an ROC curve by varying the discriminative threshold of our classifier.
- ● The ROC curve monotonically increases.

(5) [3 pts] The K-means algorithm:

- ◯ Requires the dimension of the feature space to be no bigger than the number of samples
- number of clusters
- ◯ Has the smallest value of the objective function when $K = 1$
- ◯ Converges to the global optimum if and only if the initial means are chosen as some of the samples themselves
- ● Minimizes the within class variance for a given
- ◯ None of the above

(6) [3 pts] Suppose when you are training your convolutional neural network, you find that the training loss just doesn't go down after initialization. What could you try to fix this problem?

- ● Change the network architecture
- ● Find a better model
- ● Change learning rates
- ● Normalize the inputs to the network
- ● Ensure training data is being read correctly
- ◯ Add a regularization term

(7) [3 pts] Logistic regression:

- ● Minimizes cross-entropy loss
- ○ Has a simple, closed form analytical solution
- ● Models the log-odds as a linear function
- ● Is a classification method to estimate class posterior probabilities

(8) [3 pts] Select all the true statements.

- ○ The first principal component is unique up to a sign change.
- ○ The last principal component is unique up to a sign change.
- ○ All principal components are unique up to a sign change.
- ● If some features are linearly dependent, at least one singular value is zero.
- ○ If some features are correlated, at least one singular value is zero.

(9) [3 pts] Select all the choices that make the following statement true:
In (a), the training error does not increase as (b) increases.

- ● a: K-means,
  b: number of iterations
- ○ a: Training neural nets with back propagation using *batch* gradient decent,
  b: number of iterations
- ○ a: Training neural nets with back propagation using *stochastic* gradient decent,
  b: number of iterations
- ● a: Regression Trees with square loss,
  b: depth of the tree
- ○ a: Random Forest Classifier,
  b: number of trees in the forest
- ● a: Least squares,
  b: number of features

(10) [3 pts] Neural networks:

- ○ Optimize a convex objective function
- ○ Can only be trained with stochastic gradient descent
- ● Can use a mix of different activation functions
- ● Can be made to perform well even when the number of parameters/weights is much greater than the number of data points.

Q1. [90 pts] Multiple Choice

Check the boxes for **ALL** CORRECT CHOICES. Every question should have at least one box checked. NO PARTIAL CREDIT: the set of all correct answers (only) must be checked.

(1) [3 pts] What strategies can help reduce overfitting in decision trees?

- ■ Pruning
- ☐ Make sure each leaf node is one pure class
- ■ Enforce a minimum number of samples in leaf nodes
- ■ Enforce a maximum depth for the tree

(2) [3 pts] Which of the following are true of convolutional neural networks (CNNs) for image analysis?

- ■ Filters in earlier layers tend to include edge detectors
- ■ Pooling layers reduce the spatial resolution of the image
- ☐ They have more parameters than fully-connected networks with the same number of layers and the same numbers of neurons in each layer
- ☐ A CNN can be trained for unsupervised learning tasks, whereas an ordinary neural net cannot

(3) [3 pts] Neural networks

- ☐ optimize a convex cost function
- ■ can be used for regression as well as classification
- ☐ always output values between 0 and 1
- ■ can be used in an ensemble

(4) [3 pts] Which of the following are true about generative models?

- ■ They model the joint distribution $P(\text{class} = \text{C AND sample} = \mathbf{x})$
- ■ They can be used for classification
- ☐ The perceptron is a generative model
- ■ Linear discriminant analysis is a generative model

(5) [3 pts] Lasso can be interpreted as least-squares linear regression where

- ■ weights are regularized with the ℓ_1 norm
- ☐ weights are regularized with the ℓ_2 norm
- ☐ the weights have a Gaussian prior
- ☐ the solution algorithm is simpler

(6) [3 pts] Which of the following methods can achieve zero training error on *any* linearly separable dataset?

- ■ Decision tree
- ■ Hard-margin SVM
- ☐ 15-nearest neighbors
- ■ Perceptron

(7) [3 pts] The kernel trick

- ☐ can be applied to every classification algorithm
- ☐ changes ridge regression so we solve a $d \times d$ linear system instead of an $n \times n$ system, given n sample points with d features
- ☐ is commonly used for dimensionality reduction
- ■ exploits the fact that in many learning algorithms, the weights can be written as a linear combination of input points

(8) [3 pts] Suppose we train a hard-margin linear SVM on $n > 100$ data points in $\mathbb{R}^2$, yielding a hyperplane with exactly 2 support vectors. If we add one more data point and retrain the classifier, what is the maximum possible number of support vectors for the new hyperplane (assuming the $n+1$ points are linearly separable)?

- ☐ 2
- ☐ n
- ☐ 3
- ■ $n+1$

(9) [3 pts] In latent semantic indexing, we compute a low-rank approximation to a term-document matrix. Which of the following motivate the low-rank reconstruction?

- ■ Finding documents that are related to each other, e.g. of a similar genre
- ☐ The low-rank approximation provides a lossless method for compressing an input matrix
- ■ In many applications, some principal components encode noise rather than meaningful structure
- ☐ Low-rank approximation enables discovery of nonlinear relations

(10) [3 pts] Which of the following are true about subset selection?

- ☐ Subset selection can substantially decrease the bias of support vector machines
- ■ Subset selection can reduce overfitting
- ☐ Ridge regression frequently eliminates some of the features
- ■ Finding the true best subset takes exponential time

(11) [3 pts] In neural networks, nonlinear activation functions such as sigmoid, tanh, and ReLU

- ☐ speed up the gradient calculation in backpropagation, as compared to linear units
- ■ help to learn nonlinear decision boundaries
- ☐ are applied only to the output units
- ☐ always output values between 0 and 1

(12) [3 pts] Suppose we are given data comprising points of several different classes. Each class has a different probability distribution from which the sample points are drawn. We do not have the class labels. We use k-means clustering to try to guess the classes. Which of the following circumstances would undermine its effectiveness?

- ☐ Some of the classes are not normally distributed
- ☐ The variance of each distribution is small in all directions
- ■ Each class has the same mean
- ■ You choose $k = n$, the number of sample points

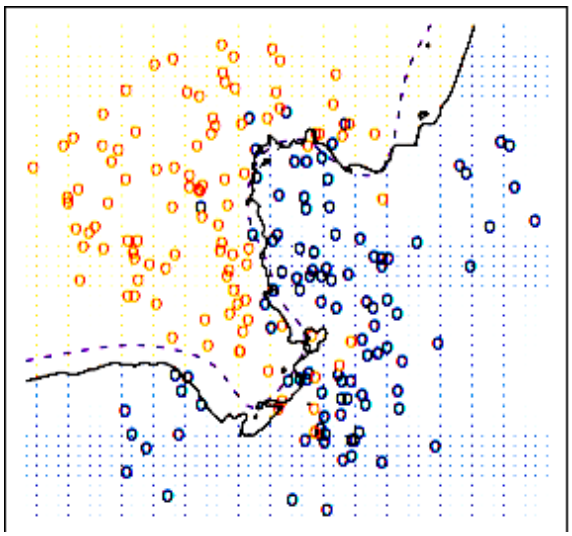
(13) [3 pts] Which of the following are true of spectral graph partitioning methods?

- ☐ They find the cut with minimum weight
- ☐ They minimize a quadratic function subject to one constraint: the partition must be balanced
- ■ They use one or more eigenvectors of the Laplacian matrix
- ☐ The Normalized Cut was invented at Stanford

(14) [3 pts] Which of the following can help to reduce overfitting in an SVM classifier?

- ■ Use of slack variables
- ☐ High-degree polynomial features
- ☐ Normalizing the data
- ☐ Setting a very low learning rate

(15) [3 pts] Which value of k in the k-nearest neighbors algorithm generates the solid decision boundary depicted here? There are only 2 classes. (Ignore the dashed line, which is the Bayes decision boundary.)



☐ $k = 1$

☐ $k = 2$

■ $k = 10$

☐ $k = 100$

(16) [3 pts] Consider one layer of weights (edges) in a convolutional neural network (CNN) for grayscale images, connecting one layer of units to the next layer of units. Which type of layer has the fewest parameters to be learned during training? (Select one.)

☐ A convolutional layer with 10 3×3 filters

☐ A convolutional layer with 8 5×5 filters

■ A max-pooling layer that reduces a 10×10 image to 5×5

☐ A fully-connected layer from 20 hidden units to 4 output units

(17) [3 pts] In the kernelized perceptron algorithm with learning rate $\epsilon = 1$, the coefficient a_i corresponding to a training example x_i represents the weight for $K(x_i, x)$. Suppose we have a two-class classification problem with $y_i \in \{1, -1\}$. If $y_i = 1$, which of the following can be true for a_i?

☐ $a_i = -1$

■ $a_i = 1$

■ $a_i = 0$

■ $a_i = 5$

(18) [3 pts] Suppose you want to split a graph G into two subgraphs. Let L be G's Laplacian matrix. Which of the following could help you find a good split?

☐ The eigenvector corresponding to the second-largest eigenvalue of L

☐ The left singular vector corresponding to the second-largest singular value of L

■ The eigenvector corresponding to the second-smallest eigenvalue of L

■ The left singular vector corresponding to the second-smallest singular value of L

(19) [3 pts] Which of the following are properties that a kernel matrix always has?

☐ Invertible

☐ All the entries are positive

☐ At least one negative eigenvalue

■ Symmetric

(20) [3 pts] How does the bias-variance decomposition of a ridge regression estimator compare with that of ordinary least squares regression? (Select one.)

- ☐ Ridge has larger bias, larger variance
- ☐ Ridge has smaller bias, larger variance
- ■ Ridge has larger bias, smaller variance
- ☐ Ridge has smaller bias, smaller variance

(21) [3 pts] Both PCA and Lasso can be used for feature selection. Which of the following statements are true?

- ■ Lasso selects a subset (not necessarily a strict subset) of the original features
- ☐ PCA and Lasso both allow you to specify how many features are chosen
- ■ PCA produces features that are linear combinations of the original features
- ☐ PCA and Lasso are the same if you use the kernel trick

(22) [3 pts] Which of the following are true about forward subset selection?

- ☐ $O(2^d)$ models must be trained during the algorithm, where d is the number of features
- ☐ It finds the subset of features that give the lowest test error
- ■ It greedily adds the feature that most improves cross-validation accuracy
- ■ Forward selection is faster than backward selection if few features are relevant to prediction

(23) [3 pts] You've just finished training a random forest for spam classification, and it is getting abnormally bad performance on your validation set, but good performance on your training set. Your implementation has no bugs. What could be causing the problem?

- ■ Your decision trees are too deep
- ■ You have too few trees in your ensemble
- ■ You are randomly sampling too many features when you choose a split
- ■ Your bagging implementation is randomly sampling sample points *without* replacement

(24) [3 pts] Consider training a decision tree given a design matrix $X = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels $y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let f_1 denote feature 1, corresponding to the first column of X, and let f_2 denote feature 2, corresponding to the second column. Which of the following splits at the root node gives the highest information gain? (Select one.)

- ☐ $f_1 > 2$
- ☐ $f_2 > 3$
- ■ $f_1 > 4$
- ☐ $f_2 > 6$

(25) [3 pts] In terms of the bias-variance decomposition, a 1-nearest neighbor classifier has _____ than a 3-nearest neighbor classifier.

- ■ higher variance
- ☐ higher bias
- ☐ lower variance
- ■ lower bias

(26) [3 pts] Which of the following are true about bagging?

- ■ In bagging, we choose random subsamples of the input points with replacement
- ☐ The main purpose of bagging is to decrease the bias of learning algorithms.
- ☐ Bagging is ineffective with logistic regression, because all of the learners learn exactly the same decision boundary
- ■ If we use decision trees that have one sample point per leaf, bagging never gives lower training error than one ordinary decision tree

(27) [3 pts] An advantage of searching for an approximate nearest neighbor, rather than the exact nearest neighbor, is that

- ☐ it sometimes makes exhaustive search much faster
- ☐ the nearest neighbor classifier is sometimes much more accurate
- ■ it sometimes makes searching in a k-d tree much faster
- ☐ you find all the points within a distance of $(1+\epsilon)r$ from the query point, where r is the distance from the query point to its nearest neighbor

(28) [3 pts] In the derivation of the spectral graph partitioning algorithm, we *relax* a combinatorial optimization problem to a continuous optimization problem. This relaxation has the following effects.

- ■ The combinatorial problem requires an exact bisection of the graph, but the continuous algorithm can produce (after rounding) partitions that aren't perfectly balanced
- ☐ The combinatorial problem requires finding eigenvectors, whereas the continuous problem requires only matrix multiplication
- ☐ The combinatorial problem cannot be modified to accommodate vertices that have different masses, whereas the continuous problem can
- ■ The combinatorial problem is NP-hard, but the continuous problem can be solved in polynomial time

(29) [3 pts] The firing rate of a neuron

- ☐ determines how strongly the dendrites of the neuron stimulate axons of neighboring neurons
- ■ is more analogous to the output of a unit in a neural net than the output voltage of the neuron
- ☐ only changes very slowly, taking a period of several seconds to make large adjustments
- ☐ can sometimes exceed 30,000 action potentials per second

(30) [3 pts] In algorithms that use the kernel trick, the Gaussian kernel

- ■ gives a regression function or predictor function that is a linear combination of Gaussians centered at the sample points
- ☐ is equivalent to lifting the d-dimensional sample points to points in a space whose dimension is exponential in d
- ■ is less prone to oscillating than polynomials, assuming the variance of the Gaussians is large
- ☐ has good properties in theory but is rarely used in practice

(31) **3 bonus points!** The following Berkeley professors were cited in this semester's lectures (possibly self-cited) for specific research contributions they made to machine learning.

- ☐ David Culler
- ■ Michael Jordan
- ■ Jitendra Malik
- ■ Leo Breiman
- ■ Anca Dragan
- ☐ Jonathan Shewchuk

Q1. [78 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(1) [3 pts] Which of the following are NP-hard problems? Let $X \in \mathbb{R}^{n \times d}$ be a design matrix, let $y \in \mathbb{R}^n$ be a vector of labels, let L be the Laplacian matrix of some n-vertex graph, and let $\mathbf{1} = [1 \quad 1 \quad \ldots \quad 1]^\top$.

- ● $\min_{\mu,y} \sum_{i=1}^{k} \sum_{y_j=i} |X_j - \mu_i|^2$ where each μ_i is the mean of sample points assigned class i
- ○ $\min_{y \in \mathbb{R}^n} \frac{1}{4} y^\top L y$ subject to $|y|^2 = n$; $\mathbf{1}^\top y = 0$
- ○ $\min_{y} \sum_{i=1}^{k} \sum_{y_j=i} |X_j - \mu_i|^2$ with each μ_i fixed
- ● $\min_{y \in \mathbb{R}^n} \frac{1}{4} y^\top L y$ subject to $\forall j, y_j \in \{-1, +1\}$; $\mathbf{1}^\top y = 0$

(2) [3 pts] Which clustering algorithms permit you to decide the number of clusters after the clustering is done?

- ○ k-means clustering
- ● a k-d tree used for divisive clustering
- ● agglomerative clustering with single linkage
- ○ spectral graph clustering with 3 eigenvectors

(3) [3 pts] For which of the following does normalizing your input features influence the predictions?

- ○ decision tree (with usual splitting method)
- ● neural network
- ● Lasso
- ● soft-margin support vector machine

(4) [3 pts] With the SVD, we write $X = UDV^\top$. For which of the following matrices are the eigenvectors the columns of U?

- ○ $X^\top X$
- ○ $X^\top X X^\top X$
- ● $XX^\top$
- ● $XX^\top XX^\top$

(5) [3 pts] Why is PCA sometimes used as a preprocessing step before regression?

- ● To reduce overfitting by removing poorly predictive dimensions.
- ● To make computation faster by reducing the dimensionality of the data.
- ○ To expose information missing from the input data.
- ○ For inference and scientific discovery, we prefer features that are not axis-aligned.

(6) [3 pts] Consider the matrix $X = \sum_{i=1}^{r} \alpha_i u_i v_i^\top$ where each α_i is a scalar and each u_i and v_i is a vector. It is possible that the rank of X might be

- ○ $r + 1$
- ● $r - 1$
- ● r
- ● 0

(7) [3 pts] Why would we use a random forest instead of a decision tree?

- ○ For lower training error.
- ● To better approximate posterior probabilities.
- ● To reduce the variance of the model.
- ○ For a model that is easier for a human to interpret.

(8) [3 pts] What tends to be true about increasing the k in k-nearest neighbors?

- [x] The decision boundary tends to get smoother.
- [x] The bias tends to increase.
- [ ] The variance tends to increase.
- [x] As the number of sample points approaches infinity (with $n/k \to \infty$), the error rate approaches less than twice the Bayes risk (assuming training and test points are drawn independently from the same distribution).

(9) [3 pts] Which of the following statements are true about the entropy of a discrete probability distribution?

- [x] It is a useful criterion for picking splits in decision trees.
- [ ] It is a convex function of the class probabilities.
- [x] It is maximized when the probability distribution is uniform.
- [ ] It is minimized when the probability distribution is uniform.

(10) [3 pts] A low-rank approximation of a matrix can be useful for

- [x] removing noise.
- [x] discovering latent categories in the data.
- [x] filling in unknown values.
- [x] matrix compression.

(11) [3 pts] Let L be the Laplacian matrix of a graph with n vertices. Let

$$\beta = \min_{\substack{y \in \mathbb{R}^n \\ \forall i, y_i \in \{-1,+1\} \\ \mathbf{1}^\top y = 0}} y^\top L y \qquad \text{and} \qquad \gamma = \min_{\substack{y \in \mathbb{R}^n \\ |y|^2 = n \\ \mathbf{1}^\top y = 0}} y^\top L y.$$

Which of the following are true for every Laplacian matrix L?

- [x] $\beta \geq \gamma$
- [ ] $\beta > \gamma$
- [ ] $\beta \leq \gamma$
- [ ] $\beta < \gamma$

(12) [3 pts] Which of the following are true about decision trees?

- [ ] They can be used only for classification.
- [ ] All the leaves must be pure.
- [ ] The tree depth never exceeds $O(\log n)$ for n sample points.
- [x] Pruning usually achieves better test accuracy than stopping early.

(13) [3 pts] Which of the following is an effective way of reducing overfitting in neural networks?

- [x] Augmenting the training data with similar synthetic examples
- [ ] Increasing the number of layers
- [x] Weight decay (i.e., ℓ_2 regularization)
- [x] Dropout

(14) [3 pts] If the VC dimension of a hypothesis class $\mathcal{H}$ is an integer $D < \infty$ (i.e., $\text{VC}(\mathcal{H}) = D$), this means

- [x] there exists some set of D points shattered by $\mathcal{H}$.
- [x] no set of $D + 1$ points is shattered by $\mathcal{H}$.
- [ ] all sets of D points are shattered by H.
- [x] $\Pi_{\mathcal{H}}(D) = 2^D$.

(15) [3 pts] Consider the minimizer w^* of the ℓ_2-regularized least squares objective $J(w) = |Xw - y|^2 + \lambda|w|^2$ with $\lambda > 0$. Which of the following are true?

- ○ $Xw^* = y$
- ○ w^* exists if and only if $X^\top X$ is nonsingular
- ○ $w^* = X^+ y$, where X^+ is the pseudoinverse of X
- ● The minimizer w^* is unique

(16) [3 pts] You are training a neural network, but the training error is high. Which of the following, if done in isolation, has a better-than-tiny chance of reducing the training error?

- ● Adding another hidden layer
- ● Adding more units to hidden layers
- ● Normalizing the input data
- ○ Training on more data

(17) [3 pts] Filters in the **late** layers of a convolutional neural network designed to classify objects in photographs likely represent

- ○ edge detectors.
- ● concepts such as "this image contains wheels."
- ● concepts such as "there is an animal."
- ○ concepts such as "Jen is flirting with Dan."

(18) [3 pts] Which of the following techniques usually speeds up the training of a sigmoid-based neural network on a classification task?

- ○ Using batch descent instead of stochastic
- ● Having a good initialization of the weights
- ○ Increasing the learning rate with every iteration
- ● Using the cross-entropy loss instead of the mean squared error

(19) [3 pts] In a soft-margin support vector machine, decreasing the slack penalty term C causes

- ○ more overfitting.
- ○ a smaller margin.
- ● less overfitting.
- ● less sensitivity to outliers.

(20) [3 pts] The shortest distance from a point z to a hyperplane $w^\top x = 0$ is

- ○ $w^\top z$
- ○ $\frac{w^\top z}{|w|^2}$
- ● $\frac{w^\top z}{|w|}$
- ○ $|w| \cdot |z|$

(21) [3 pts] The Bayes decision rule

- ● does the best a classifier can do, in expectation
- ● chooses the class with the greatest posterior probability, if we use the 0-1 risk function
- ○ can be computed exactly from a large sample
- ● minimizes the risk functional

(22) [3 pts] Which of the following are techniques commonly used in training neural nets?

- ○ linear programming
- ○ Newton's method
- ● backpropagation
- ● cross-validation

(23) [3 pts] Which of these statements about learning theory are correct?

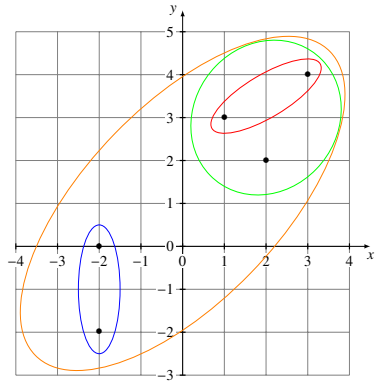
- ● The VC dimension of halfplanes is 3.
- ○ The VC dimension of halfspaces in 3D is ∞.
- ○ For a fixed set of training points, the more dichotomies Π we have, the higher the probability that the training error is close to the true risk.
- ● For a fixed hypothesis class $\mathcal{H}$, the more training points we have, the higher the probability that the training error is close to the true risk.

(24) [3 pts] Which of the following statements are true for a design matrix $X \in \mathbb{R}^{n \times d}$ with $d > n$? (The rows are n sample points and the columns represent d features.)

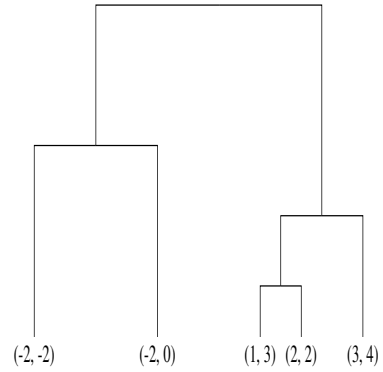
- ○ Least-squares linear regression computes the weights $w = (X^\top X)^{-1} X^\top y$.
- ○ The sample points are linearly separable.
- ○ X has exactly $d - n$ eigenvectors with eigenvalue zero.
- ● At least one principal component direction is orthogonal to a hyperplane that contains all the sample points.

(25) [3 pts] Which of the following visuals accurately represent the clustering produced by greedy agglomerative hierarchical clustering with centroid linkage on the set of feature vectors {(-2, -2), (-2, 0), (1, 3), (2, 2), (3, 4)}?

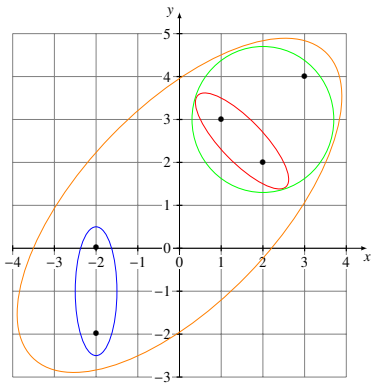
○



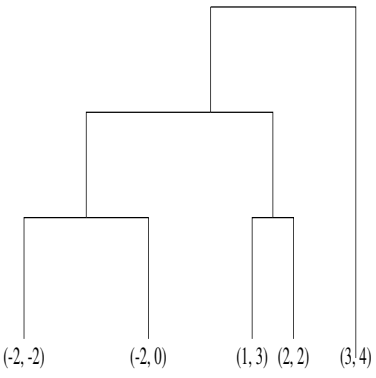
●



●



○



(26) [3 pts] Which of the following statements is true about the standard k-means clustering algorithm?

- ○ The random partition initialization method usually outperforms the Forgy method.
- ○ It is computationally infeasible to find the optimal clustering of $n = 15$ points in $k = 3$ clusters.
- ● After a sufficiently large number of iterations, the clusters will stop changing.
- ○ You can use the metric $d(x, y) = \frac{x \cdot y}{|x| \cdot |y|}$.

Q1. [64 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(1) [4 pts] Which of the following are true for the ***k*-nearest neighbor** (*k*-NN) algorithm?

- ● A: *k*-NN can be used for both classification and regression.
- ● B: As k increases, the bias usually increases.
- ○ C: The decision boundary looks smoother with smaller values of k.
- ○ D: As k increases, the variance usually increases.

(2) [4 pts] Let X be a matrix with **singular value decomposition** $X = U\Sigma V^\top$. Which of the following are true for all X?

- ● A: $\text{rank}(X) = \text{rank}(\Sigma)$.
- ○ B: If all the singular values are unique, then the SVD is unique.
- ● C: The first column of V is an eigenvector of $X^\top X$.
- ● D: The singular values and the eigenvalues of $X^\top X$ are the same.

A is correct because the number of non-zero singular values is equal to the rank. B is incorrect because you could change both $U \to -U, V \to -V$. C is correct because the SVD and eigendecomposition of $X^\top X$ is $V\Sigma^2 V^\top$. D is correct as $X^\top X$ is positive semidefinite, so the eigenvalues can't be negative.

(3) [4 pts] Lasso (with a fictitious dimension), random forests, and principal component analysis (PCA) **all** ...

- ● A: can be used for dimensionality reduction or feature subset selection
- ○ B: compute linear transformations of the input features
- ○ C: are supervised learning techniques
- ● D: are *translation invariant*: changing the origin of the coordinate system (i.e., translating all the training and test data together) does not change the predictions or the principal component directions

Option B is incorrect because random forests don't compute linear transformations. Option C is incorrect because PCA is unsupervised.

(4) [4 pts] Suppose your training set for two-class classification in one dimension ($d = 1$; $x_i \in \mathbb{R}$) contains three sample points: point $x_1 = 3$ with label $y_1 = 1$, point $x_2 = 1$ with label $y_2 = 1$, and point $x_3 = -1$ with label $y_3 = -1$. What are the values of w and b given by a **hard-margin SVM**?

- ○ A: $w = 1, b = 1$
- ○ B: $w = 0, b = 1$
- ● C: $w = 1, b = 0$
- ○ D: $w = \infty, b = 0$

(5) [4 pts] Use the same training set as part (d). What is the value of w and b given by **logistic regression** (with no regularization)?

- ○ A: $w = 1, b = 1$
- ○ B: $w = 0, b = 1$
- ○ C: $w = 1, b = 0$
- ● D: $w = \infty, b = 0$

(6) [4 pts] Below are some choices you might make while training a neural network. Select all of the options that will generally make it **more difficult** for your network to achieve high accuracy on the test data.

● A: Initializing the weights to all zeros

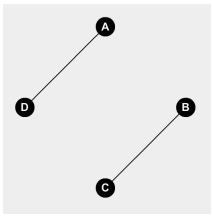
○ C: Using momentum

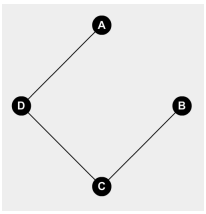
● B: Normalizing the training data but leaving the test data unchanged

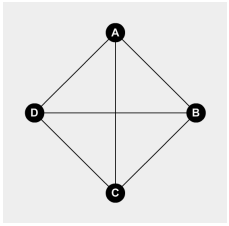
○ D: Reshuffling the training data at the beginning of each epoch

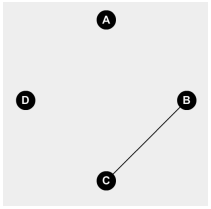
A) Initializing weights with zeros makes it impossible to learn. B) Mean and standard deviation should be computed on the training set and then used to standardize the validation and test sets, so that the distributions are matched for each set. C) This describes momentum and will generally help training. D) This is best practice.

(7) [4 pts] To the left of each graph below is a number. Select the choices for which the number is the multiplicity of the eigenvalue zero in the **Laplacian matrix** of the graph.

○ A: 1 

○ C: 2 

● B: 1 

○ D: 4 

The multiplicity is equal to the number of connected components in the graph.

(8) [4 pts] Given the spectral graph clustering optimization problem

$$\begin{aligned} &\text{Find } y \text{ that minimizes } y^\top L y \\ &\text{subject to} \quad y^\top y = n \\ &\text{and} \quad \mathbf{1}^\top y = 0, \end{aligned}$$

which of the following optimization problems produce a vector y that leads to **the same sweep cut** as the optimization problem above? M is a diagonal mass matrix with different masses on the diagonal.

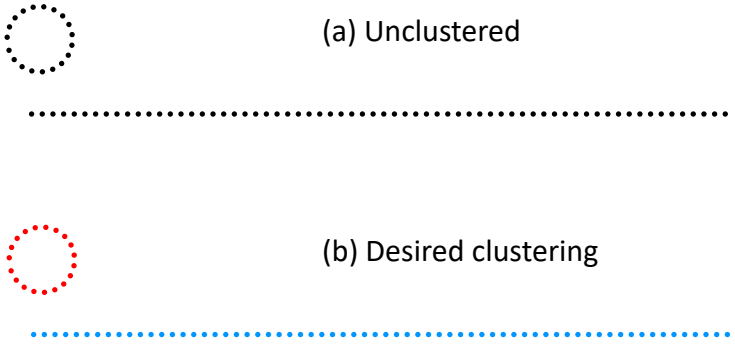
● A: Minimize $y^\top L y$ subject to $y^\top y = 1$ and $\mathbf{1}^\top y = 0$

○ B: Minimize $y^\top L y$ subject to $\forall i,\ y_i = 1$ or $y_i = -1$ and $\mathbf{1}^\top y = 0$

● C: Minimize $y^\top L y / (y^\top y)$ subject to $\mathbf{1}^\top y = 0$

○ D: Minimize $y^\top L y$ subject to $y^\top M y = 1$ and $\mathbf{1}^\top M y = 0$

(9) [4 pts] Which of the following methods will cluster the data in panel (a) of the figure below into the two clusters (red circle and blue horizontal line) shown in panel (b)? Every dot in the circle and the line is a data point. In all the options that involve hierarchical clustering, the algorithm is run until we obtain two clusters.



○ A: Hierarchical agglomerative clustering with Euclidean distance and complete linkage

● B: Hierarchical agglomerative clustering with Euclidean distance and single linkage

○ C: Hierarchical agglomerative clustering with Euclidean distance and centroid linkage

○ D: k-means clustering with $k = 2$

Single linkage uses the minimum distance between two clusters as a metric for merging clusters. Since the two clusters are densely packed with points and the minimum distance between the two clusters is greater than the within-cluster distances between points, single linkage doesn't link the circle to the line until the very end.

The other three methods will all join some of the points at the left end of the line with the circle, before they are joined with the right end of the line.

(10) [4 pts] Which of the following statement(s) about **kernels** are true?

● A: The dimension of the lifted feature vectors $\Phi(\cdot)$, whose inner products the kernel function computes, can be infinite.

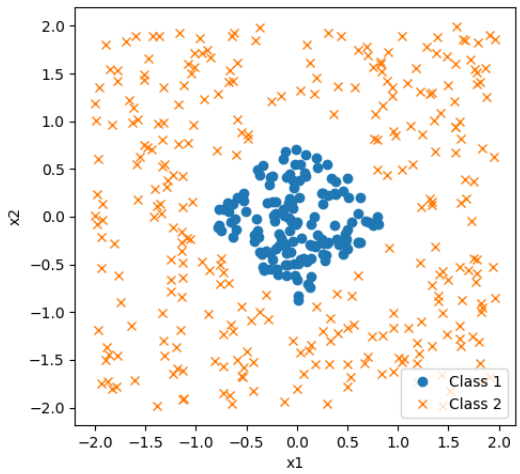
○ B: For any desired lifting $\Phi(x)$, we can design a kernel function $k(x, z)$ that will evaluate $\Phi(x)^\top \Phi(z)$ more quickly than explicitly computing $\Phi(x)$ and $\Phi(z)$.

● C: The kernel trick, when it is applicable, speeds up a learning algorithm if the number of sample points is substantially less than the dimension of the (lifted) feature space.

● D: If the raw feature vectors x, y are of dimension 2, then $k(x, y) = x_1^2 y_1^2 + x_2^2 y_2^2$ is a valid kernel.

A is correct; consider the Gaussian kernel from lecture. B is wrong; most liftings don't lead to super-fast kernels. Just some special ones do. C is correct, straight from lecture. Though in this case, the dual algorithm is faster than the primal whether you use a fancy kernel or not. D is correct because $k(x, y)$ is inner product of $\Phi(x) = [x_1^2 \quad x_2^2]^\top$ and $\Phi(y) = [y_1^2 \quad y_2^2]^\top$.

(11) [4 pts] We want to use a **decision tree** to classify the training points depicted. Which of the following decision tree classifiers is capable of giving 100% accuracy on the training data with **four splits or fewer**?



○ A: A standard decision tree with axis-aligned splits

○ B: Using PCA to reduce the training data to one dimension, then applying a standard decision tree

● C: A decision tree with multivariate linear splits

● D: Appending a new feature $|x_1| + |x_2|$ to each sample point x, then applying a standard decision tree

A standard decision tree will need (substantially) more than four splits. PCA to 1D will make it even harder. However, four non-axis-aligned multivariate linear splits suffice to cut the diamond out of the center. Finally, adding the L_1 norm feature lets us perfectly classify the data with a single split that cuts off the top of the pyramid.

(12) [4 pts] Which of the following are true about **principal components analysis** (PCA)?

○ A: The principal components are eigenvectors of the centered data matrix.

● B: The principal components are right singular vectors of the centered data matrix.

● C: The principal components are eigenvectors of the sample covariance matrix.

● D: The principal components are right singular vectors of the sample covariance matrix.

The first three follow directly from definitions. The last is because the covariance matrix is symmetric, so the singular vectors are the eigenvectors.

(13) [4 pts] Suppose we are doing **ordinary least-squares linear regression** with a fictitious dimension. Which of the following changes can **never** make the cost function's value on the **training data** smaller?

● A: Discard the fictitious dimension (i.e., don't append a 1 to every sample point).

○ B: Append quadratic features to each sample point.

● C: Project the sample points onto a lower-dimensional subspace with PCA (without changing the labels) and perform regression on the projected points.

● D: Center the design matrix (so each feature has mean zero).

A: Correct. Discarding the fictitious dimension forces the linear regression function to be zero at the origin, which may increase the cost function but can never decrease it.

B: Incorrect. Added quadratic features often help to fit the data better.

C: Correct. Regular OLS is at least as expressive. Projecting the points may incrase the cost function but can never decrease it. Centering features doesn't matter so WLOG assume X has centered features. If the full SVD is $X = U\Sigma V^\top$, then projecting onto a k-dimensional subspace gives $X_k = U\Sigma_k V^\top$. If w_k is a solution for the PCA-projected OLS, we can take $w = Vz$ where z is the first k elements of $V^\top w_k$ with the rest zero, and get $X_k w_k = Xw$.

D: Correct. Since we're using a fictitious dimension, translating the points does not affect the cost of the optimal regression function (which translates with the points).

(14) [4 pts] Which of the following are true about **principal components analysis** (PCA)? Assume that no two eigenvectors of the sample covariance matrix have the same eigenvalue.

● A: Appending a 1 to the end of every sample point doesn't change the results of performing PCA (except that the useful principal component vectors have an extra 0 at the end, and there's one extra useless component with eigenvalue zero).

● B: If you use PCA to project d-dimensional points down to j principal coordinates, and then you run PCA again to project those j-dimensional coordinates down to k principal coordinates, with $d > j > k$, you always get the same result as if you had just used PCA to project the d-dimensional points directly down to k principle coordinates.

○ C: If you perform an arbitrary rigid rotation of the sample points as a group in feature space before performing PCA, the principal component directions do not change.

● D: If you perform an arbitrary rigid rotation of the sample points as a group in feature space before performing PCA, the largest eigenvalue of the sample covariance matrix does not change.

Appending an extra dimension with the same values introduces no variance in the extra dimension, so PCA will ignore that dimension. PCA discards the eigenvector directions associated with the largest eigenvalues; as the eigenvectors are mutually orthogonal, this does not affect the variance in the surviving dimensions, so your results depend solely on how many directions you discard. Rotating the sample points rotates the principal components, but it doesn't change the variance along each of those (rotated) component directions.

(15) [4 pts] Consider running a single iteration of **AdaBoost** on three sample points, starting with uniform weights on the sample points. All the ground truth labels and predictions are either $+1$ or -1. In the table below, some values have been omitted. Which of the following statements can we say **with certainty**?

	True Label	Classifier Prediction	Initial Weight	Updated Weight
X_1	-1	-1	$1/3$	?
X_2	?	$+1$	$1/3$	$\sqrt{2}/3$
X_3	?	?	$1/3$	$\sqrt{2}/6$

● A: X_1's updated weight is $\sqrt{2}/6$

○ B: X_3's classifier prediction is -1

● C: X_2 is misclassified

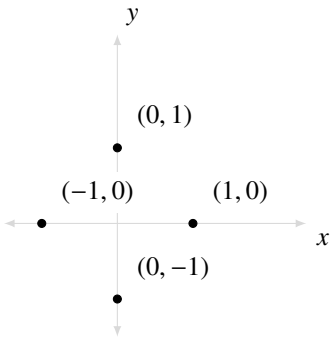
○ D: X_3 is misclassified

In the AdaBoost algorithm, all correctly classified points have their weights changed by the same multiplicative factor. Since we observe two different updated weights, we know one of x_2 or x_3 is correctly classified, and the other is misclassified. Since x_1 is correctly classified, the error rate is err $= 1/3$. As the error rate is less than $1/2$, the weights of correctly classified points will decrease and the weights of misclassified points will increase. Hence, X_2 is misclassified and X_3 is correctly classified. As X_1 is correctly classified, it has the same updated weight as X_3. But we can't tell what X_3's classifier prediction is; only that it is correctly classified.

As an aside, we can confirm the multipliers used for reweighting of misclassified and correctly classified points (in that order):

$$\sqrt{\frac{\text{err}}{1-\text{err}}} = \sqrt{\frac{2/3}{1/3}} = \sqrt{2} \qquad \sqrt{\frac{1-\text{err}}{\text{err}}} = \sqrt{\frac{1/3}{2/3}} = \frac{\sqrt{2}}{2}$$

(16) [4 pts] Consider running the **hierarchical agglomerative clustering** algorithm on the following set of four points in $\mathbb{R}^2$, breaking ties arbitrarily. If we stop when only two clusters remain, which of the following linkage methods *ensures* the resulting clusters are balanced (each have two sample points)? Select all that apply.



- [x] A: Complete linkage
- [ ] B: Single linkage
- [x] C: Centroid linkage
- [x] D: Average linkage

Under each of these linkage methods, we know that two adjacent points along a side of the rhombus will first be fused together. Without loss of generality, assume these are the points $(0, 1)$ and $(1, 0)$. Treating these as a cluster, the average, maximum and centroid distances to each of the two remaining points are all larger that the distance between the two points themselves. Therefore, complete linkage, single linkage and average linkage all result in balanced clusters. On there other hand, single linkage does not, since, for example, $(1, 0)$ and $(0, -1)$ have the same distance as $(-1, 0)$ and $(0, -1)$.

Q1. [60 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(a) [4 pts] Which of the following conditions could serve as a sensible stopping condition while building a decision tree?

- ○ A: Stop if you find the validation error is decreasing as the tree grows
- ● C: Don't split a treenode whose depth exceeds a specified threshold
- ○ B: Don't split a treenode that has an equal number of sample points from each class
- ○ D: Don't split a treenode if the split would cause a large reduction in the weighted average entropy

A: It is sensible to stop when validation error is increasing, but not when it is decreasing. B: This is the kind of node for which splitting is most likely to be effective, so this is not a sensible stopping condition. C: It is a common practice to limit the maximum depth of the tree. D: This is the kind of node for which splitting is most effective, so this is not a sensible stopping condition.

(b) [4 pts] Let classifier A be a random forest. Let classifier B be an ensemble of decision trees with bagging—identical to classifier A except that we do not limit the splits in each treenode to a subset of the features; at every treenode, the very best split among all d features is chosen. Which statements are true?

- ○ A: After training, all the trees in classifier B must be identical
- ● C: Classifier B will tend to have higher variance than Classifier A
- ○ B: Classifier B will tend to have higher bias than Classifier A
- ● D: Classifier B will tend to have higher training accuracy than Classifier A

A is incorrect because we still use bagging. **B** is incorrect because limiting the splits, as a random forest does, tends to increase the bias. **C** is correct because randomly limiting the splits, as a random forest does, tends to decrease the ensemble variance—this is what motivated random forests in the first place! **D** is correct because reduced bias increases training accuracy.

(c) [4 pts] We are using an ensemble of decision trees for a classification problem, with bagging. We notice that the decision trees look too similar. We would like to build a more diverse set of learners. What are possible ways to accomplish that?

- ○ A: Increase the size of each random subsample
- ○ C: Apply normalization to the design matrix first
- ● B: Decrease the size of each random subsample
- ○ D: Sample without replacement

(d) [4 pts] Suppose we have a feature map Φ and a kernel function $k(X_i, X_j) = \Phi(X_i) \cdot \Phi(X_j)$. Select the true statements about kernels.

- ○ A: If there are n sample points of dimension d, it takes $O(nd)$ time to compute the kernel matrix
- ● B: The kernel trick implies we do not compute $\Phi(X_i)$ explicitly for any sample point X_i
- ○ C: For every possible feature map $\Phi : \mathbb{R}^d \to \mathbb{R}^D$ you could imagine, there is a way to compute $k(X_i, X_j)$ in $O(d)$ time
- ● D: Running times of kernel algorithms do not depend on the dimension D of the feature space $\Phi(\cdot)$

(e) [4 pts] Which of the following are benefits of using the backpropogation algorithm to compute gradients?

- ○ A: Its running time is linear in the total number of units (neurons) in the network
- ● B: It can be applied to any arithmetic function (assuming the directed computation graph has no cycles and we evaluate the gradient at a point where the gradient exists)

● C: Compared to naive gradient computation, it improves the speed of each iteration of gradient descent by eliminating repeated computations of the same subproblem

○ D: Compared to naive gradient computation, it reduces the number of iterations required to get close to a local minimum, by protecting against sigmoid unit saturation (vanishing gradients)

A: Its running time is linear in the number of edges in the network, not the number of units. The number of edges could be quadratic in the number of units.

B: The backpropogation algorithm requires computing gradients in topological order given the reversed computational graph. Thus, any graph that can be topologically sorted (any DAG) can be backpropogated over.

C: Yes; that is its purpose.

D: Backpropagation doesn't change the gradient or descent step taken by each iteration; it just speeds up computing the gradient. Therefore, the number of iterations doesn't change and the vanishing gradient problem is not diminished.

(f) [4 pts] Which of the following are benefits of using convolutional neural networks—as opposed to fully connected ones—for image recognition tasks?

○ A: The ability to express a wider variety of more complicated functions of the input features

○ B: Fewer model architecture hyperparameters for the designer to select

● C: Enables the network to more easily learn and recognize features regardless of their position in the image

● D: Typically requires less data to train well

In principal, fully connected networks offer greater expressivity and model capacity as they have many more weights; so option A has it backwards. Since CNNs share weights, the same features can be learned and recognized at different positions. Many fewer parameters often implies less data is required to train a reasonable model. CNNs involve more architecture parameters (kernel size, stride, dilation, padding, pooling, etc.).

(g) [4 pts] Select the correct statements about principal component analysis (PCA).

● A: PCA is a method of dimensionality reduction

● B: If we select only one direction (a one-dimensional subspace) to represent the data, the sample variance of the projected points is zero if and only if the original sample points are all identical

● C: The orthogonal projection of a point x onto a unit direction vector w is $(x^\top w)w$

○ D: If we select only one direction (a one-dimensional subspace) to represent the data, PCA chooses the eigenvector of the sample covariance matrix that corresponds to the least eigenvalue

A and C are standard knowledge that we told you in lecture. B is true because PCA chooses the variance-maximizing direction; if there are two sample points that are not identical to each other, there is a direction with strictly positive variance. D is false because we want the eigenvector associated with the *greatest* eigenvalue.

(h) [4 pts] Consider a **centered** design matrix $X \in \mathbb{R}^{n\times d}$, where $X_i \in \mathbb{R}^d$ is the i-th sample point (a column vector) and $X_i^\top$ is the i-th row of X. Which of these optimization problems is a correct formulation of finding the first principal component v?

● A: Subject to $\|v\|_2 = 1$, find $v \in \mathbb{R}^d$ that minimizes $\sum_{i=1}^n \|X_i - vv^\top X_i\|_2^2$

○ B: Find $v \in \mathbb{R}^d$ that minimizes $\dfrac{v^\top X^\top X v}{\|v\|_2^2}$

○ C: Subject to $\|v\|_2 = 1$, find $v \in \mathbb{R}^d$ that maximizes $\sum_{i=1}^n \|X_i - vv^\top X_i\|_2^2$

● D: Find $v \in \mathbb{R}^d$ that maximizes $\dfrac{v^\top X^\top X v}{\|v\|_2^2}$

(i) [4 pts] Consider a **centered** design matrix $X \in \mathbb{R}^{n\times d}$ and its singular value decomposition $X = UDV^\top$. X has d principal components, which are found by principal components analysis (PCA). Which statements are correct?

● A: The row space of X (if not trivial) is spanned by some of the principal components

● B: The null space of X (if not trivial) is spanned by some of the principal components

● C: The principal components are all right singular vectors of X

● D: The matrix UD lists the principal coordinates of every sample point in X

A, B: The row space is spanned by the principal components with nonzero singular values, and the null space is spanned by the principal components with zero singular values. C is true because the right singular vectors of X are the eigenvectors of $X^\top X$. D is true because $UD = XV$.

(j) [4 pts] Select the correct statements about the Fiedler vector.

○ A: The Fiedler vector is the eigenvector of the Laplacian matrix that is associated with the smallest eigenvalue

● B: The Fiedler vector always satisfies the balance constraint (as written as an equation)

○ C: The Fiedler vector is a solution to the unrelaxed, NP-hard optimization problem

● D: The *sweep cut* is a spectral graph partitioning technique that tries $n - 1$ different cuts (in an n-vertex graph) and picks one of them.

Option A is incorrect because it's associated with the second-smallest eigenvalue. Option B is correct because $\mathbf{1}$ is always an eigenvector of the Laplacian matrix, eigenvectors are orthogonal to each other, and the balance condition is $\mathbf{1}^T y = 0$. Option C is incorrect; the Fiedler vector is a solution to the relaxed problem, but probably not the unrelaxed one. Option D is a fact.

(k) [4 pts] Consider the optimization problem of finding $q \in \mathbb{R}^r$ that minimizes $\|y - XPq\|_2^2 + \lambda\|q\|_2^2$, where $\lambda > 0$, $X \in \mathbb{R}^{n \times d}$ is a design matrix, $y \in \mathbb{R}^n$ is a vector of labels, and $P \in \mathbb{R}^{d \times r}$ is an arbitrary matrix with $r < d$. This problem has one unique solution. Which of the following is that one unique solution?

- ● A: $q = (P^\top X^\top XP + \lambda I)^{-1} P^\top X^\top y$
- ○ B: $q = (X^\top P^\top PX + \lambda I)^{-1} P^\top X^\top y$
- ○ C: $q = (P^\top X^\top XP + \lambda I)^{-1} X^\top y$
- ○ D: $q = (X^\top P^\top PX + \lambda I)^{-1} X^\top y$

(l) [4 pts] Select the correct statements about AdaBoost.

- ● A: "Ada" stands for "adaptive," as the meta-learner adapts to the performance of its learners
- ○ B: AdaBoost works best with support vector machines
- ● C: At test/classification time, AdaBoost computes a weighted sum of predictions
- ○ D: AdaBoost can transform any set of classifiers to a classifier with better training accuracy

A and C are basic. B is false: linear classifiers are less well suited to ensembling than most nonlinear classifiers are. D is false: if the weak classifiers all have 50% accuracy, then AdaBoost can't build a reliable meta-learner, even for just the training points.

(m) [4 pts] Select the correct statements about AdaBoost.

- ○ A: When we go from iteration t to iteration $t + 1$, the weight of sample point X_i is increased if the majority of the t learners misclassify X_i.
- ○ B: Unlike with decision trees, two data points that are identical ($X_i = X_j$) but have different labels ($y_i \neq y_j$) can be classified correctly by Adaboost.
- ● C: AdaBoost can benefit from a learner that has only 5% training accuracy.
- ○ D: If you train enough learners and every learner achieves at least 51% validation accuracy, Adaboost can always achieve 100% validation accuracy.

A: The change in weight of a data point is only determined by the most recent learner at time t, not any of the previous learners. B: Even though AdaBoost is an ensemble, it only has one output for a given X_i. C: AdaBoost treats a learner with 5% training accuracy by flipping its output so it has 95% training accuracy; then it can benefit very much! D: Would be true for training accuracy, but of course it's false for validation accuracy!

(n) [4 pts] In which of the following cases should you prefer ***k*-nearest neighbors over *k*-means clustering**? For all the four options, you have access to images $X_1, X_2, \ldots, X_n \in \mathbb{R}^d$.

- ○ A: You do **not** have access to labels. You want to find out if any of the images are very different from the rest, i.e., are outliers.
- ○ B: You have access to labels $y_1, y_2, \ldots, y_n$ telling us whether image i is a cat or a dog. You want to find out whether the distribution of cats is unimodal or bimodal. You already know that the distribution of cats either has either one or two modes, but that's all you know about the distribution.
- ● C: You have access to labels $y_1, y_2, \ldots, y_n$ telling us whether image i is a cat or a dog. You want to find out whether a new image z is a cat or a dog.
- ● D: You have access to labels $y_1, y_2, \ldots, y_n$ telling us whether image i is a cat or a dog. Given a new image z, you want to approximate the posterior probability of z being a cat and the posterior probability of z being a dog.

A: You don't have access to labels, so you cannot use k-nearest neighbors.

B: In order to do this, you would first filter out all the dogs, so you're left with cat images only. Then you perform k-means clustering twice, once with $k = 1$ and again with $k = 2$. If the total distance between the data points and the cluster centers are markedly lower in the second case than the first, that would suggest that the data is bimodal.

C: You want to perform classification, so you must use k-NN. k-means is a clustering algorithm and not a classification algorithm.

D: In this case, you must use k-NN. Instead of finding the plurality class in the k points closest to z, you find the class histogram of those points.

(o) [4 pts] Select the correct statements about the k-nearest neighbor classifier.

- ● A: For exact nearest neighbor search in a very high-dimensional feature space, generally it is faster to use exhaustive search than to use a k-d tree.
- ● B: When using a k-d tree, approximate nearest neighbor search is sometimes substantially faster than exact nearest neighbor search.
- ○ C: When using exhaustive search, approximate nearest neighbor search is sometimes substantially faster than exact nearest neighbor search.
- ○ D: Order-k Voronoi diagrams are widely used in practice for k-nearest neighbor search (with $k > 1$) in a two-dimensional feature space.

A: In very high-dimensional spaces, k-d trees with exact search generally don't succeed in pruning many points away, so you're doing exhaustive search anyway, only with the greater (by a constant factor) overhead of searching a tree data structure too.

B: Yes; sometimes by a factor of up to 100.

C: No; you don't gain any speedup at all.

D: No; they take up $O(k^2n)$ space, and there is a lack of publicly available software for these difficult computations.

Q1. [60 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(a) [4 pts] What is true of convolutional neural networks (CNNs)?

● A: Learned convolutional masks can act as edge detectors or line detectors.

● B: A convolutional layer of connections from a layer of m hidden units to a layer of m' hidden units has fewer weights than a fully-connected layer of connections from a layer of m hidden units to a layer of m' hidden units.

○ C: Let X be a 4×4 image that is fed through an average-pooling layer with 2×2 masks, producing a 2×2 output layer Y. Then for every unit/pixel X_{ij} and every unit/pixel $Y_{k\ell}$, the partial derivative $\partial Y_{k\ell}/\partial X_{ij}$ is equal to $1/4$.

● D: Some research on CNNs made a strong enough impact to win the Alan M. Turing Award.

A is mentioned in the lecture on CNNs.
B just says that convolutional layers aren't fully connected.
C is false. Although every non-zero entry of the matrix is equal to $\frac{1}{4}$, there are also many zeros within the $\frac{\partial Y}{\partial X}$ tensor stemming from the fact that some Y values do not depend at all on some other X values.
D is mentioned in the lecture on CNNs.

(b) [4 pts] What is true of regression algorithms?

○ A: All regression problems can be solved by solving a system of linear equations.

○ B: Least squares regression can be derived from maximum likelihood estimation if we assume that the sample points have a multivariate normal distribution.

○ C: Adding a feature with no predictive value to least squares regression is unwise because it increases the bias of the method.

● D: *Weighted* least squares regression can be derived from maximum likelihood estimation if we assume that some points' labels are noisier than others.

A: Logistic regression and Lasso are counterexamples. B: least squares regression assumes that the label noise is modeled by a Gaussian, not the sample points. C: no, it will not increase the bias, though it might decrease it. However, it is unwise, as it might increase the variance. D is true, weighted least squares regression can be motivated when different points have different noise variances.

(c) [4 pts] What is true of kernels?

○ A: Kernel algorithms require you to solve a linear system with a $D \times D$ matrix, where D is the length of the lifted sample points $\Phi(X_i)$.

○ B: Neural networks can be kernelized because there is always an optimal weight vector w that is a linear combination of the sample points.

○ C: Fitting and evaluating a kernel ridge regression model requires access only to the kernel matrix K and not the training points X_i nor $\Phi(X_i)$.

● D: The kernel perceptron algorithm with a Gaussian kernel returns a classifier similar to a smoothed version of k-nearest neighbor classification.

A: If that were true, kernel methods would be so slow nobody would use them.
B: Haha; the weight vector is generally not even the same length as the sample points.
C: To evaluate the decision/regression function on a test point z, we need access to the training points so we can compute kernels $k(X_i, z)$.
D: As stated on page 92 of Lecture 16.

(d) [4 pts] Which of the following statements are true of the Rayleigh quotient $Q(X, w) = w^\top X^\top X w/(w^\top w)$ for an arbitrary matrix $X \in \mathbb{R}^{n \times d}$ and vector $w \in \mathbb{R}^d$?

● A: $Q(X, w) \geq 0$ for all X and w.

● B: $Q(X, w) \leq \sigma^2_{\max}(X)$ for all X and w, where $\sigma_{\max}(X)$ is the greatest singular value of X.

● C: $Q(X, w)$ is maximized when w is the eigenvector of $X^\top X$ corresponding to the greatest eigenvalue.

○ D: $Q(X, w)$ is maximized when w is the eigenvector of $X^\top X$ corresponding to the smallest eigenvalue.

A is true because $w^\top X^\top X w = \|Xw\|^2$ cannot be negative, nor can $w^\top w = \|w\|^2$. C was stated in lecture. For the maximizing w, the length of Xw is $\sigma_{\min}$ times the length of w, hence $\|Xw\|^2$ is $\sigma^2_{\min}$ times $\|w\|^2$, so C implies B.

(e) [4 pts] Given $X \in \mathbb{R}^{n \times d}$ and $y \in \mathbb{R}^n$, consider solving the normal equations $X^\top X w = X^\top y$ for $w \in \mathbb{R}^d$. Let the SVD of X be $X = UDV^\top$. Let X^+ denote the Moore–Penrose pseudoinverse of a matrix X. Which of the following statements are certain to be true, for any values of X and y?

○ A: $w = (X^\top X)^{-1} X^\top y$ is a solution of the normal equations.

● B: $w = X^+ y$ is a solution of the normal equations.

● C: $w = VD^+ U^\top y$ is a solution of the normal equations.

○ D: Every solution w to the normal equations is a solution to $Xw = y$.

A is incorrect because $(X^\top X)$ might be singular, thus not invertible. But the pseudoinverse is always defined, and B is correct. C follows from B by writing out a formula for X^+. D is incorrect; $Xw = y$ does not always have a solution.

(f) [4 pts] Which of the following would be a reasonable cost function (by the criteria discussed in lecture) for choosing splits in a decision tree for two-class classification, where p is the fraction of points in Class C in a specified treenode?

- ● A: $-p \log_2 p - (1-p)\log_2(1-p)$
- ○ B: $p \log_2 p + (1-p)\log_2(1-p)$
- ○ C: $0.5 - |p - 0.5|$
- ● D: $p(1-p)$

A is the entropy (for two classes), which works well. B is minus entropy, which is convex and counterproductive; it favors bad splits. C is not strictly concave and is similar to the percent misclassified, and so is poor for the reasons given in class for the latter. D is the Gini impurity, also briefly mentioned in class, which works well.

(g) [4 pts] Which of the following are advantages to using AdaBoost with short trees (say, depth 4) over random forests with an equal number of tall trees (refined until the leaves are pure)?

- ○ A: AdaBoost is more robust against overfitting outliers in the training data.
- ● B: AdaBoost is faster to train.
- ○ C: AdaBoost is better at reducing variance than a random forest.
- ● D: AdaBoost is better at reducing bias than a random forest.

A: AdaBoost is worse at handling outliers because of the exponential loss function and the fact that it generally reaches 100% training accuracy. B is correct because AdaBoost uses shorter trees. C is wrong because bagging and randomizing the split choices are designed to reduce variance, whereas AdaBoost doesn't actively try to reduce variance and empirically sometimes overfits in practice. D is correct because AdaBoost is explicitly designed to reduce bias, and usually can reach 100% training accuracy given enough trees, whereas random forests tend to increase bias.

(h) [4 pts] What is true of k-nearest neighbor classifiers in a Euclidean space?

- ● A: A 1-NN classifier is more likely to overfit and less likely to underfit than a 10-NN classifier.
- ● B: In the special case of a two-dimensional feature space, it is possible to preprocess a data set so that nearest neighbor queries can be answered in $O(\log n)$ time.
- ● C: Sometimes finding k approximate nearest neighbors can be done much faster than finding the k exact nearest neighbors.
- ○ D: The k-d tree algorithm for nearest neighbor search can speed up 1-NN, but it doesn't work well for k-NN.

A: Of course. B: Yes, build a planar point location data structure on a Voronoi diagram. C: Of course. D: Wrong; k-NN is nearly as easy as 1-NN.

(i) [4 pts] What is true of bagging? (Note: not in a random forest; just bagging alone.)

- ● A: Bagging without replacement is more likely to overfit than bagging with replacement.
- ○ B: Bagging involves using different learning algorithms on different subsamples of the training set.
- ○ C: Bagging is often used with decision trees because it helps increase their training accuracy.
- ● D: Even with bagging, sometimes decision trees still end up looking very similar.

A: Bagging without replacement tends to make your subsamples more alike, which makes your trees more alike, rather than "decorrelating" them, so you're more likely to overfit. In the extreme case, if $n' = n$, all your trees will be identical. B: No, it's always the same learning algorithm. If you're using different learning algorithms, there's generally no reason to subsample. C: A single decision tree can achieve 100% training accuracy if we refine it enough and there are no two coincident training points from different classes, whereas bagging might not achieve 100% training accuracy. Bagging is used to reduce variance, not to reduce bias or increase training accuracy. D: True, if there are a few features that dominate/are very strong predictors. That's what motivated the invention of random forests.

(j) [4 pts] What is true of k-means clustering?

○ A: k-means is a supervised learning algorithm.

○ B: k-means clustering always converges to the same solution regardless of how clusters are initialized.

● C: Increasing k can never increase the optimal value of the k-means cost function.

● D: The k-medoids algorithm with the ℓ_1 distance is less sensitive to outliers than standard k-means with the Euclidean distance.

A is false; k-means clustering does not require any input labels. B is false; k-means is sensitive to initialization and converges to a local minimum which might not be the global minimum. C is true; adding more clusters can only decrease your loss for k-means (proven in Discussion Section 13). D is true. Penalizing ℓ_1 distances is generally less sensitive to outliers than penalizing ℓ_2 (Euclidean) distances.

(k) [4 pts] Which of the following statements about AdaBoost is true?

○ A: AdaBoost is a natural good fit with the 1-nearest neighbor classifier.

● B: AdaBoost can give you a rough estimate of the posterior probability that a test point is in the predicted class.

○ C: AdaBoost trains multiple weak learners that classify test points by equal majority vote.

● D: AdaBoost with an ensemble of soft-margin linear SVM classifiers allows the metalearner to learn nonlinear decision boundaries.

A: No. The 1-NN algorithm always gets 100% training accuracy (if there are no coinciding points from different classes), so what is there to boost? And how would you use the AdaBoost's weights in 1-NN?
B: True, from the continous output of the metalearner; see the lecture.
C: The weak learners are weighted based on their error rate.
D: True.

(l) [4 pts] You have n training points, each with d features. You try two different algorithms for training a decision tree.

1. The standard decision tree with single-feature (axis-aligned) splits, as covered in lecture.
2. A *randomized* decision tree—like a tree in a random forest, but there is only one tree and we don't use bagging. At each internal node of the tree, we randomly select m of the d features, and we choose the best split from among those m features.

Suppose you train both trees permitting no treenode to have depth greater than h. Upon completion, you find that in each tree, at least half its leaves are at depth h. What are the running times for training the standard decision tree and the randomized one, respectively? (Select exactly one option.)

○ A: $\Theta(dn2^h)$; $\Theta(mn2^h)$

○ B: $\Theta(dn2^h)$; $\Theta(dn2^h)$

● C: $\Theta(dnh)$; $\Theta(mnh)$

○ D: $\Theta(dnh)$; $\Theta(dnh)$

A training points X_i participates only in the treenodes in the path from the root to the leaf node that contains X_i, and there are at most $h + 1$ such treenodes. If we sum over all n training points, there are $\Theta(nh)$ point-treenode pairs (consisting of a point that participates in a treenode).

During training (tree construction), the cost of processing a treenode in the standard tree is d times the number of training points that participate in the treenode. Summing this over treenodes, we have $\Theta(nh)$ point-treenode pairs to process each requiring d time, so the overall running time is in $\Theta(dnh)$.

The cost of processing a treenode in the randomized tree is m times the number of training points that participate, so the overall running time is in $\Theta(mnh)$.

(m) [4 pts] Suppose you have a fixed dataset and want to train a decision tree for two-class classification. If you increase the maximum depth of the decision tree, which of the following are possible effects?

● A: The test accuracy goes up.

○ B: The training accuracy goes down.

○ C: The number of pure leaves is reduced.

● D: The time to classify a test point increases.

A: this is possible, and is generally the goal of making a decision tree more complex.
B: accuracy on the training set cannot go down.
C: pure leaves in the short tree remain pure leaves in the large tree.
D: adding layers adds more nodes for the training algorithm to traverse.

(n) [4 pts] What is true of the shatter function $\Pi_H(n)$, where n is the number of training points?

● A: It either grows polynomially with n or is 2^n for all n, but it can never be in between those extremes.

● B: We use it to compute an estimate of how many training points we need to ensure we choose a hypothesis with nearly optimal risk with high probability.

● C: For a linear classifier in $\mathbb{R}^d$, the shatter function grows polynomially with n.

○ D: For a power set classifier, the shatter function grows polynomially with n.

All of these come from the lecture.

(o) [4 pts] What is true of human neurology?

○ A: The output of a unit in an artificial neural network is roughly analogous to the voltage of an axon at the synapse.

● B: A connection in an artificial neural network is roughly analogous to a synapse in the brain.

○ C: The brain is made of general-purpose neurons, each of which could be trained to do any job that neurons do.

● D: The visual cortex has neurons primarily devoted to detecting lines or edges.

A is wrong—the output is analogous to the firing rate, not the voltage. B and D are empirically true. C is very wrong. The brain has many functional regions that are specialized to do very different things. A neuron in the visual cortex can't be trained to do emotion.